TEAM 2 PRESENTATION HANDBOOK

# Historical Stock Analysis

From the first row of the dataset to every claim in the 10-slide presentation

**THE CENTRAL QUESTION**

Does the diversification visible in a global stock dataset survive closer inspection across technology groups and crisis periods?

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Prepared for the 27 August 2026 halfway presentation

# Read this in layers

You do not need a finance background. The guide deliberately moves from intuition to method to presentation language.

- First pass: read pages 3-16 to understand the story, dataset, and basic measurements.
- Second pass: study the three findings and the chart redesign on pages 18-25.
- Presentation pass: rehearse the slide-by-slide pages and speaker scripts on pages 29-33.
- Final pass: use the Q&A, claim boundaries, glossary, and number sheet on pages 34-40.

**One-sentence version**

The project finds that diversification looks thinner in two situations: selected chip/AI stocks dominate broader technology growth, and regional markets move together more during sudden panic, especially in 2020.

# The whole project in one minute

This is a data-visualization coursework project, not a trading strategy or a price forecast.

### What we asked

Is apparent diversification actually strong, or does it weaken when technology gains concentrate and markets enter a crisis?

### What we did

Used daily adjusted prices to compute returns, indexed growth, regional correlations, volatility, and relative volume in Python and R.

### What we found

The selected chip/AI basket gained about 549% versus 168%; regional correlation peaked at 0.539 in 2020; panic months had the largest volatility spikes.

### What we can claim

These patterns are present in this dataset for the selected stocks and periods. They do not prove causality, predict the future, or describe every company or market.

### Presentation logic

Question -> trustworthy data -> three testable hypotheses -> three evidence slides -> chart redesign -> conclusion and dashboard next step.

SECTION 1

# Start with the data

Understand what one row means, which fields matter, why adjusted prices are used, and why raw price comparison would be misleading.

# What is in stock_data.csv?

The dataset is a daily panel: one ticker on one trading day.

| Fact | Verified value | Why it matters |
| --- | --- | --- |
| Rows | 230,111 | Many ticker-days, not 230,111 companies |
| Columns | 8 | Date, ticker, prices, and volume |
| Tickers | 49 | Large companies across six market labels |
| Date range | 2006-01-02 to 2026-02-20 | About 20 years; the last row is not today |
| Missing fields | 0 | No missing-value imputation was needed |
| Duplicate Date-Ticker | 0 | Each ticker-day key is unique |
| Zero-volume rows | 1,706 | A real caveat, concentrated in AZN |

**Source caveat**

The original Kaggle dataset page was removed by its uploader. We still name the original URL, but we also disclose that the page is no longer available.

**Time caveat**

The dataset ends on 20 February 2026. Never call its final observation current, live, or today's price.

# How to read one row

A row records prices and trading activity for one company on one date.

| Field | Plain-language meaning | Use in this project |
|---|---|---|
| Date | Trading date | Ordering, filtering, monthly bins |
| Ticker | Short exchange code for a stock | Group and region identity |
| Open | Price near the start of the session | Descriptive only |
| High | Highest price that day | Descriptive only |
| Low | Lowest price that day | Descriptive only |
| Close | Split-adjusted closing price in this file | Useful, but not preferred for total return |
| Adj Close | Close additionally adjusted for dividends | Main price used for returns and growth |
| Volume | Number of shares traded | Panic-period supporting measure |

Example interpretation: Date = 2026-02-20, Ticker = AAPL, Open = 258.97, Adj Close = 264.58, Volume = 42,044,900 means Apple opened near 258.97, ended at an adjusted value of 264.58, and approximately 42 million shares were recorded as traded that day.

**Important**

A price number has meaning only within its ticker and currency. A Samsung price quoted in Korean won cannot be placed beside a U.S. dollar price and treated as directly comparable.

# Complete does not mean perfectly comparable

The file has no missing fields, but it still contains structural limitations.

- Currencies differ. U.S. stocks are mostly in dollars, Korean stocks in won, and European stocks in local currencies.
- Listing dates differ. Some tickers have shorter histories, so a common comparison start date must be checked.
- Trading calendars differ. Exchanges close on different holidays, which can create gaps in cross-market joins.
- Market coverage is uneven. 39 of 49 tickers are U.S.-listed; Paris and Saudi Arabia have one company each.
- Volume is share count, not dollar value. Stock splits change share counts and make direct company-to-company volume comparisons unsafe.
- Zero volume is recorded on 1,706 rows. That is a quality caveat even though the field is not missing.

**Good analysis habit**

Ask two different questions: Is the value present? Is the value comparable for the question I want to answer?

# Why we use Adjusted Close

The goal is to measure investor experience more faithfully across time.

### Stock split

A company can divide each old share into several new shares. The per-share price drops mechanically even though total ownership value does not. In this dataset, Close already reflects historical splits.

### Dividend

A company can pay cash to shareholders. The price typically falls by roughly the dividend amount, even though the shareholder received value. Adj Close accounts for this effect.

**Therefore, the project uses Adj Close for daily returns and indexed growth. This avoids treating a split or dividend adjustment as if it were an ordinary market gain or loss.**

### Safe presentation line

Adjusted Close is the most suitable field here because it accounts for historical splits and dividends, allowing more meaningful return and growth comparisons through time.

SECTION 2

# Build the measurement toolkit

Learn daily return, indexed growth, correlation, volatility, equal weighting, statistical significance, and what each number can and cannot say.

# Daily return: today's proportional change

Daily return removes currency units and expresses movement as a proportion.

$$return_t = (Adj\ Close_t / Adj\ Close_{(t-1)}) - 1$$

Worked example: if adjusted close rises from 100 to 103, the daily return is (103 / 100) - 1 = 0.03 = +3%. If it falls from 100 to 97, the return is -3%.

Why it helps: a 3% move means the same proportional change whether the stock is quoted at 50 dollars, 500 euros, or 50,000 won.

### First row per ticker

A return needs a previous observation. Therefore, the first row of each ticker has no daily return and is excluded from return-based summaries.

### What return does not solve

Currency-free return comparison does not remove differences in sector, country, company size, listing history, or data coverage.

# Index-to-100: give every stock the same starting line

Indexing is used for cumulative growth charts when raw price levels differ.

$$\text{Indexed value\_t} = 100 \times \text{Adj Close\_t} / \text{Adj Close\_start}$$

If a series starts at 100 and ends at 250, it gained 150%. If it ends at 80, it lost 20%. The displayed units are index points, not dollars or rupees.

### Why 100?

It is an intuitive neutral starting value. Every line begins together, so the chart compares relative growth after the chosen date.

### Critical condition

Every plotted ticker must have a valid observation on the rebase date. The date must be selected only after checking availability.

### Do not confuse these

An index ending at 649 means the value is 6.49 times the start, which is a gain of 549%, not a gain of 649%.

# Correlation: do two return series move together?

Pearson correlation ranges from -1 to +1 and has no unit.

| Correlation | Plain meaning | Diversification implication |
| --- | --- | --- |
| Near +1 | Usually rise and fall together | Less protection from each other |
| Near 0 | Little linear co-movement | Potentially more independent behavior |
| Near -1 | Often move in opposite directions | Potential offset, though rare here |

For five regions there are 10 unique pairs. The one-stock method computes a correlation matrix for each period, extracts those 10 cross-region correlations, and averages them.

**Interpret 0.539 carefully**

It means moderate positive co-movement on average during the 2020 window. It does not mean prices were equal, every day moved together, or 53.9% of losses were caused by the crisis.

**Correlation is not causation**

Two markets can move together because of a common shock. The statistic alone cannot prove why they moved together.

# Volatility: how large are the movements?

The project uses two related volatility summaries, both based on daily returns.

### Regional period volatility

For each ticker or regional return series in a period, calculate the standard deviation of daily returns, then average. Higher values mean returns are more spread out.

### Monthly absolute return

Within each ticker-month, average the absolute daily returns. Then average those ticker-month values equally across the available tickers.

Absolute return ignores direction. A -5% day and a +5% day both contribute 5% to the monthly absolute-return measure. That is intentional because volatility concerns movement size, not whether the market rose or fell.

**Why equal weighting?**

Each ticker contributes one ticker-month average. This reduces the chance that a market with many listed companies dominates the global monthly line simply because it has more rows.

**Language for the slide**

October 2008 and March 2020 had the largest average absolute daily movements in the dataset, not the largest cumulative losses.

# Volume: supporting evidence, not the headline

Volume records share count and is normalized to each ticker's own calm-period baseline.

**Volume ratio = period mean volume / ticker's 2015-2017 mean volume**

A ratio of 2.7x means average recorded share volume in the period was 2.7 times that ticker's own calm-period average before the ratios were averaged.

- One-stock regional analysis: calm 1.0x, 2008 about 2.7x, 2020 about 1.7x, 2022 about 0.9x.
- Widened baskets: calm 1.0x, 2008 about 2.99x, 2020 about 1.74x, 2022 about 0.87x.
- The conclusion is qualitative: panic periods show stronger volume activity than the 2022 drawdown in these data.

**Why the caveat?**

Share count is affected by stock splits and company size. It is not dollar trading activity, so we avoid direct raw-volume comparisons across companies.

# Welch's t-test: is the average-return gap larger than noise?

The extra-credit analysis checks H1 using pooled daily returns from the two selected groups.

Null idea: the two groups have the same mean daily return. Alternative idea: their means differ. Welch's test is suitable when the groups may have unequal variances and sample sizes.

| Group | Mean daily return | 95% confidence interval |
|---|---|---|
| Chip / AI | 0.26% | 0.19% to 0.34% |
| Broader tech | 0.14% | 0.09% to 0.19% |

### Result

Welch t = 2.69 and p = 0.0073. Under this test, the observed average-return difference would be unlikely if the two population means were truly equal.

### What p = 0.0073 does not mean

It is not the probability that H1 is true. It is not an effect size. It does not prove AI caused the returns, and it does not validate every company associated with AI.

**The day-to-day mean gap is modest, but many small differences can compound into a large cumulative gap over three years.**

SECTION 3

# The three hypotheses

Each hypothesis tests one piece of the same diversification story. Understand the claim before looking at the evidence.

# One umbrella question, three testable claims

A hypothesis is a specific statement that can be supported, rejected, or qualified with data.

### H1 - AI-rally concentration

The selected chip/AI group will outperform the selected broader-tech group during the post-2023 rally.

### H2 - Regional diversification in crises

Average regional return correlations will be higher in crisis periods than in the calm 2015-2017 baseline.

### H3 - Volatility and volume in panic

Volatility and trading volume will rise during panic periods relative to the calm baseline.

**H1 tests concentration within large-cap technology. H2 and H3 test whether apparent geographic diversification weakens during market stress.**

# Groups and periods were defined before interpretation

Visible definitions prevent the audience from imagining a wider universe than the project actually tested.

| Role | Members |
|---|---|
| Chip / AI group | NVDA, AVGO, TSM, ASML, AMD, MU, LRCX |
| Broader-tech group | AAPL, MSFT, GOOGL, AMZN, META, NFLX, ORCL, CSCO |
| One-stock regions | JPM, 005930.KS, 0700.HK, NOVN.SW, MC.PA |
| Widened US | JPM, AAPL, XOM |
| Widened Korea | 005930.KS, 000660.KS |
| Widened Hong Kong | 0700.HK, 0939.HK, 1398.HK |
| Widened Switzerland | NOVN.SW, ROG.SW |
| Widened Paris | MC.PA only |

| Period | Exact window | Interpretive label |
|---|---|---|
| Calm | 2015-01-01 to 2017-12-31 | Reference baseline |
| 2008 crisis | 2008-09-01 to 2009-03-31 | Global banking shock |
| 2020 crash | 2020-02-15 to 2020-04-30 | Sudden pandemic selloff |
| 2022 drawdown | 2022-01-01 to 2022-12-31 | Slower inflation/rate shock |

SECTION 4

# Read the evidence

Now connect each chart to its method, numbers, conclusion, and limitations. This is the core of the presentation.

# The selected chip/AI group pulled decisively ahead

Slide 5 uses indexed group averages from 3 January 2023 through 20 February 2026.

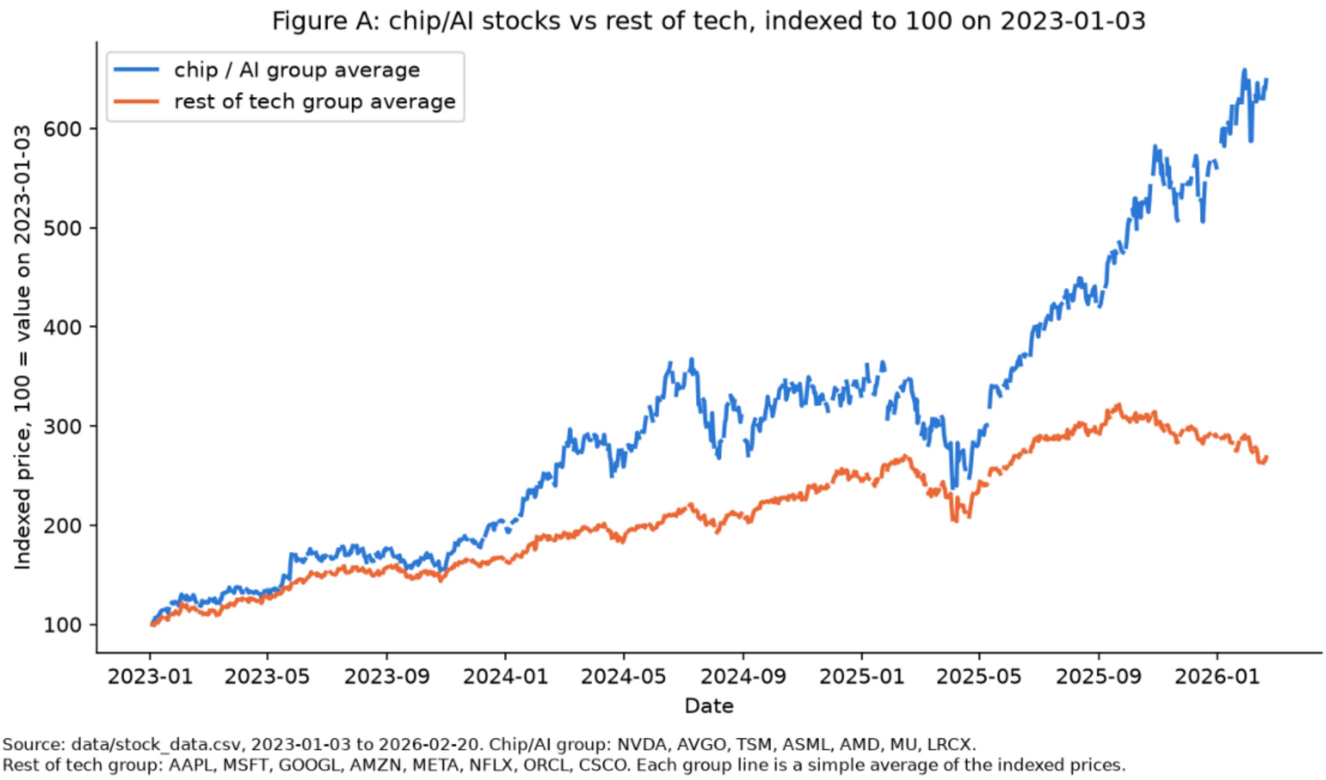



Figure A from Assignment 3. Every ticker is rebased to 100, then the indexed series are averaged within each selected group.

### Read the endpoint correctly

The chip/AI line ends near 649, a gain of about 549%. The broader-tech line ends near 268, a gain of about 168%.

# What H1 means and what it does not mean

The chart and statistical test answer related but different questions.

### Cumulative-growth chart

Shows how one unit invested at the common start would have grown on average within each selected basket. The separation persists and widens over time.

### Daily-return test

Shows the chip/AI group's pooled mean daily return was 0.26% versus 0.14%, with Welch $p = 0.0073$.

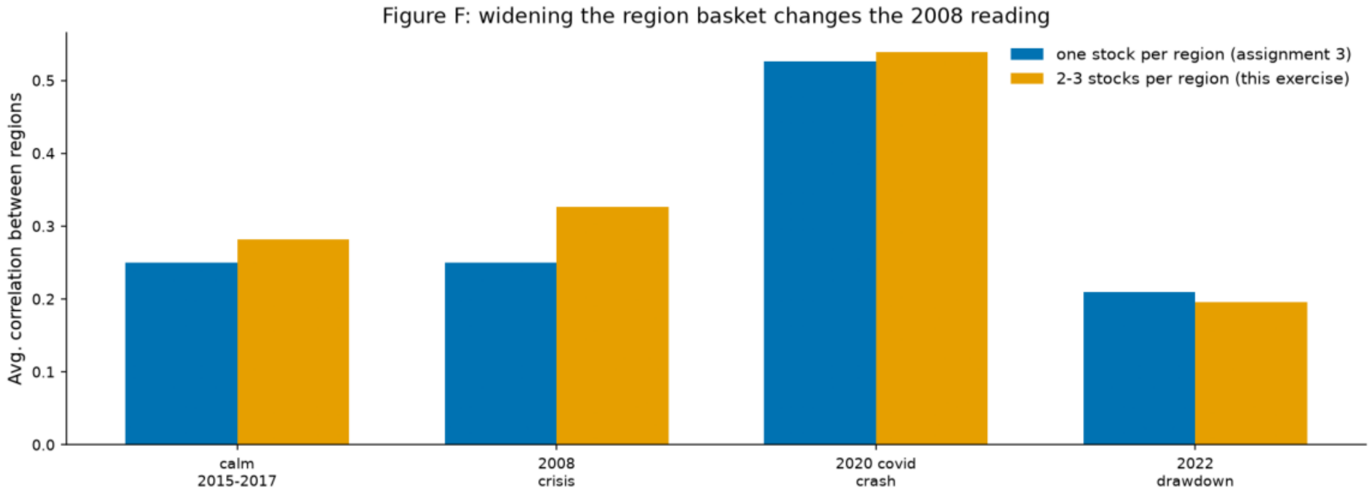
- Supported claim: the selected seven chip/AI names outperformed the selected eight broader-tech names over this period.
- Not supported: every AI-linked company outperformed, the whole AI economy is narrow, or AI caused the return difference.
- Selection matters. A different definition of AI exposure or a different starting date could change the magnitude.
- The group line is a simple average of indexed values, so each ticker receives equal weight rather than market-cap weight.

### Thirty-second explanation

We normalized every stock to the same starting value, averaged within two fixed baskets, and found a much larger cumulative gain for the chip/AI basket. A separate Welch test on daily returns points in the same direction, so H1 is supported for these selected groups.

# Widening the regional basket improves the 2008 reading

Slide 6 shows that method choice matters, especially when one company is asked to represent an entire region.




Blue uses one stock per region. Orange averages 2-3 available stocks per region, except Paris, which remains one stock.

| Period | One stock | Widened basket |
|---|---|---|
| Calm | 0.250 | 0.282 |
| 2008 | 0.250 | 0.327 |
| 2020 | 0.529 | 0.539 |
| 2022 | 0.209 | 0.196 |

# Why the 2020 conclusion is strongest

Higher positive correlation means regional returns supplied less independent movement.

- Calm widened-basket average: 0.282.
- 2008 widened-basket average: 0.327, higher than calm but not dramatically so.
- 2020 widened-basket average: 0.539, the clear maximum.
- 2022 widened-basket average: 0.196, below calm despite being a market drawdown.

### Conclusion

H2 is strongly supported for the sudden 2020 crash and directionally supported for 2008. The 2022 result shows that not every downturn produces the same correlation pattern.

### Why 2008 changed

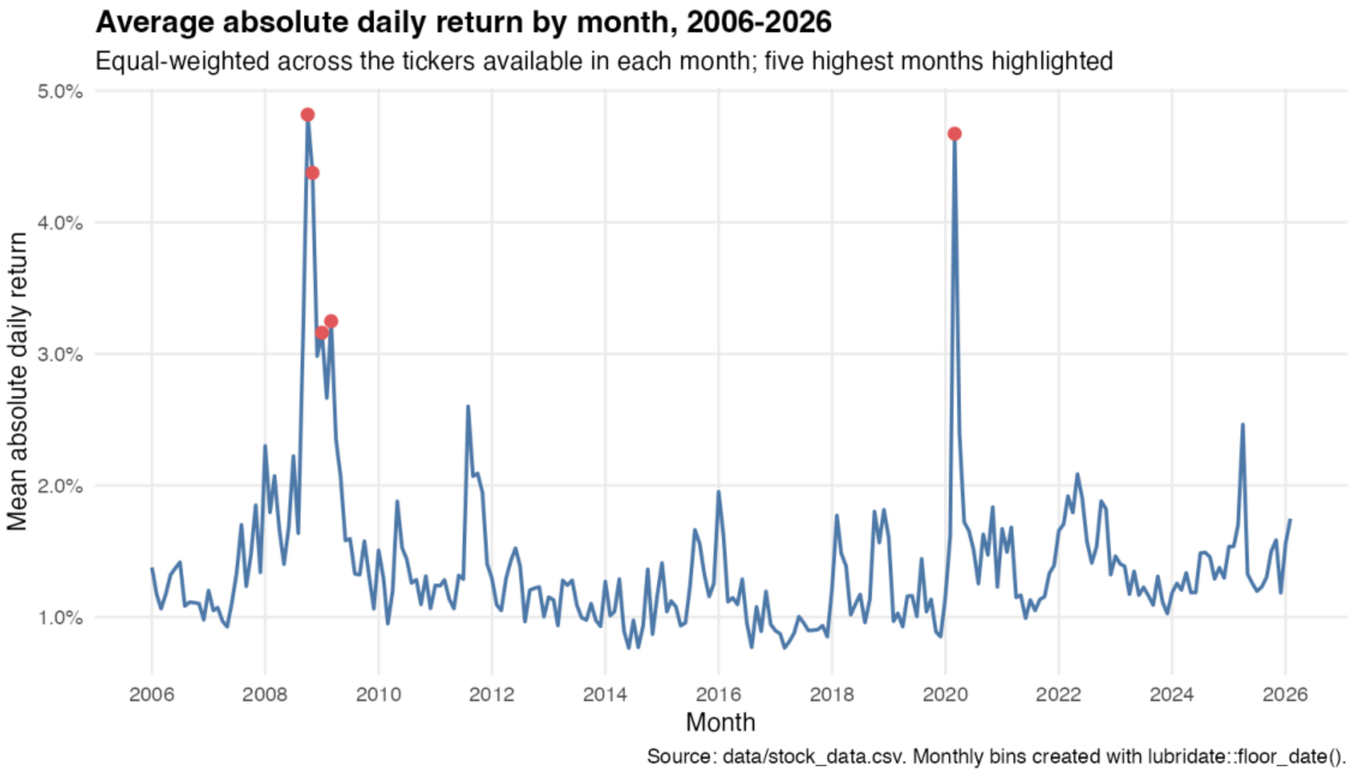
The one-stock method used JPM for the U.S. JPM had major company-specific activity around the Bear Stearns acquisition. A broader U.S. basket reduces dependence on that single company.

### Remaining limitation

Paris still has one ticker; Korea is technology-heavy; Switzerland is pharmaceutical-heavy. The result is qualified evidence, not a universal statement about regional markets.

# Panic months produced the largest volatility spikes

Slide 7 uses an independent R monthly aggregation over the full 2006-2026 dataset.




Mean absolute daily return is first averaged within each ticker-month, then equal-weighted across tickers available that month.

### Two headline months

October 2008 reached 4.82%; March 2020 reached 4.67%. November 2008 was also extremely high at 4.38%.

# H3 joins two kinds of panic evidence

Volatility is the stronger result; share-count volume is supporting evidence.

| Period | One-stock volatility | Volume vs calm |
|---|---|---|
| Calm | 0.015 | 1.0x |
| 2008 | 0.046 | 2.7x |
| 2020 | 0.036 | 1.7x |
| 2022 | 0.020 | 0.9x |

The regional-period analysis says daily returns were much more dispersed in 2008 and 2020 than in calm years. The monthly R analysis independently locates the largest average absolute movements in the same broad crisis episodes.

### Conclusion

H3 is supported in this dataset: panic periods show strong volatility and volume spikes, while 2022 looks more like a slower drawdown than a synchronized panic.

### Precise language

Say 'volatility and volume spiked during panic periods in this dataset.' Do not say 'high volume causes crashes' or 'every crisis has high correlation.'

SECTION 5

# Understand the workflow

Know what Python and R contributed, how data changed shape, and why the analysis is reproducible rather than hand-built.

# Python explored, tested, and visualized

The Python notebooks carry the main hypothesis workflow.

- Load stock_data.csv and parse Date as a date field.
- Sort by Ticker and Date so each return uses the correct previous observation.
- Create Daily Return with groupby(Ticker) and Adj Close percentage change.
- Pivot returns into date-by-ticker matrices for correlations and regional comparisons.
- Filter fixed periods and ticker groups, compute summary metrics, and create charts.
- Use SciPy for 95% confidence intervals and Welch's t-test.
- Use Matplotlib, Seaborn, Plotly, and Altair for static and interactive visual work.

**Reproducibility**

The numbers and figures come from executable notebooks or scripts. Generated figures and tables are stored beside the assignment that owns them.

**Data stays read-only**

The shared source file is not edited. Derived results belong in assignment-specific figures and tables folders.

# R replicated and reshaped the evidence

R is not a decorative second language; it independently rebuilds core summaries and prepares dashboard-friendly data.

- Assignment 4 repeats the exploratory analysis in tidyverse, data.table, lubridate, reshape2, base R, ggplot2, and lattice.
- Assignment 5 Part 2 parses dates with ymd(), creates month bins with floor_date(), and compares wide versus long formats.
- dplyr and data.table produce the same grouped market-return results, which acts as a consistency check.
- The monthly volatility chart uses long-form data and equal weighting across available ticker-months.

### Wide format

One row per month with separate AAPL, MSFT, NVDA, and AVGO columns. Easy to read as a table.

### Long format

One Month column, one Ticker column, and one value column. Easier for ggplot color mapping and Power BI/Tableau filters.

### Dashboard connection

Long form lets one Ticker field drive color, grouping, legends, and filters consistently across charts.

# The calculation pipeline

This is the shortest technical explanation that still shows control of the method.

| | | |
|---|---|---|
| 1 | **Inspect** | Check shape, fields, dates, missing values, duplicates, and coverage. |
| 2 | **Prepare** | Parse dates, sort within ticker, derive market labels, and calculate returns. |
| 3 | **Normalize** | Use returns or index-to-100 rather than mixed-currency raw prices. |
| 4 | **Define** | Fix ticker groups and crisis windows before reading the outcome. |
| 5 | **Measure** | Compute indexed growth, correlations, volatility, volume ratios, and the Welch test. |
| 6 | **Visualize** | Use direct takeaways, accessible colors, appropriate scales, and source notes. |
| 7 | **Qualify** | State unequal coverage, basket narrowness, volume limits, and no causal claim. |

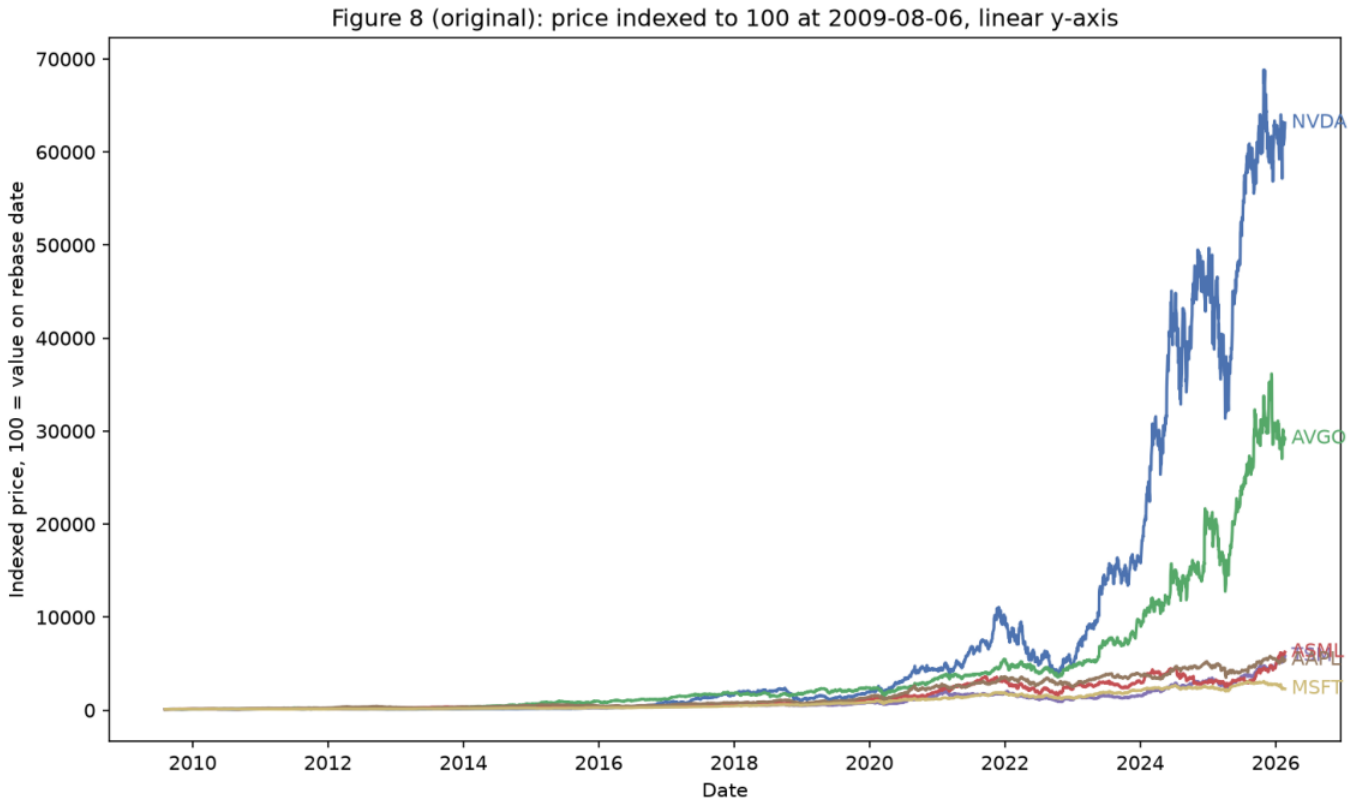
SECTION 6

# Design evolution

The before-and-after chart teaches a visualization principle: the scale should match the comparison the audience needs to make.

# Before: the linear scale hides most growth stories

The source data and index-to-100 calculation are valid, but the visual encoding makes comparison difficult.

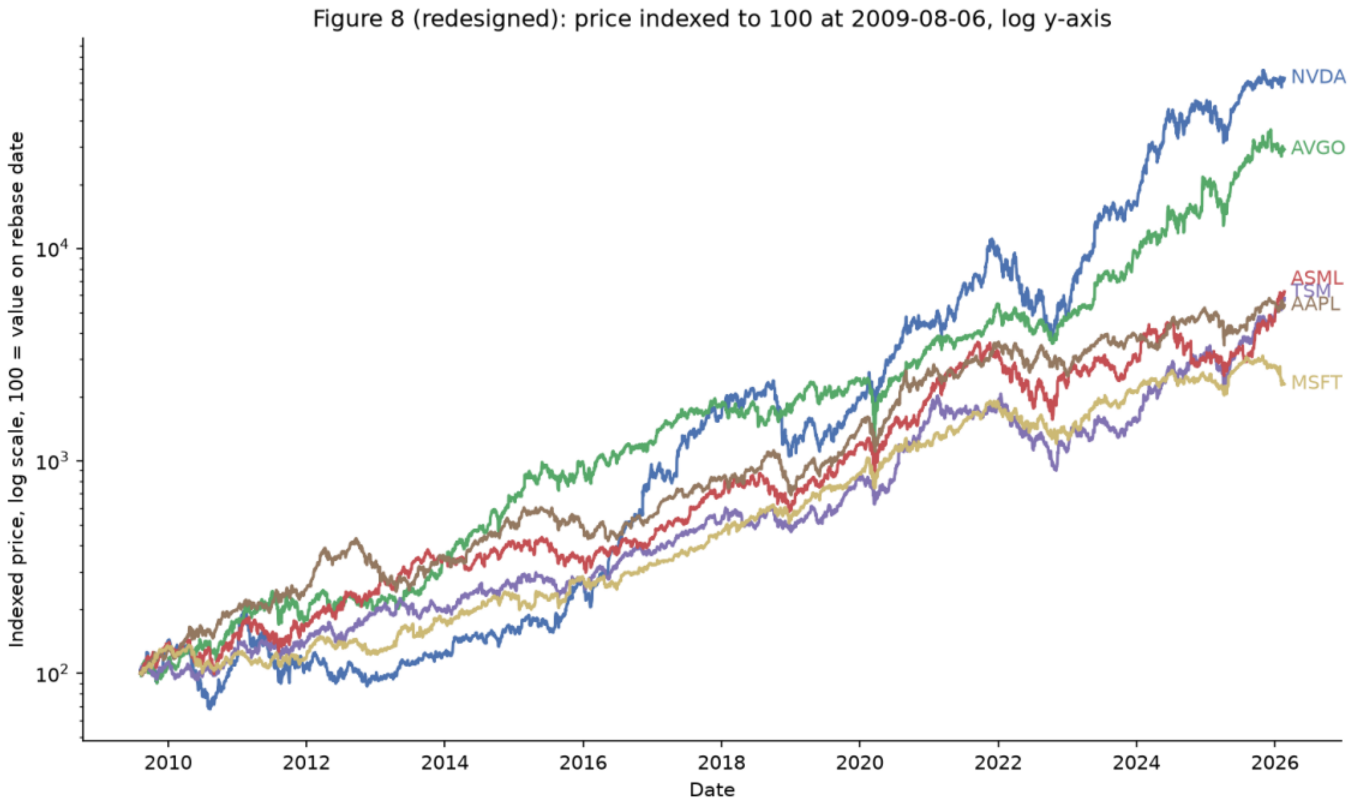



Original indexed-growth chart on a linear y-axis.

- NVDA and AVGO occupy most of the vertical range.
- ASML, TSM, AAPL, and MSFT appear compressed near the bottom.
- Several direct labels overlap near the right edge.
- Equal vertical distance represents equal index-point change, not equal percentage change.

# After: a log scale supports proportional comparison

The data and rebasing do not change; only the y-axis encoding and label placement change.




Redesigned chart with a logarithmic y-axis and separated end labels.

- Equal vertical distance now represents equal multiplicative or proportional change.
- All six growth paths remain visible across the 16-year period.
- Direct labels reduce eye movement between the lines and a legend.
- Top and right spines are removed, increasing the useful data-ink ratio.

# When a log axis is appropriate

A log scale is useful, but it must be explained clearly and cannot be used blindly.

### Use it when

All values are positive, the range spans large multiples, and the question is proportional growth rather than absolute point differences.

### Avoid or rethink it when

Values include zero or negatives, exact absolute gaps matter, or the audience could misread the axis without visible explanation.

Simple example: the move from 100 to 200 is the same proportional doubling as 1,000 to 2,000. A log axis gives both doublings the same vertical distance.

### Design-slide script

The original chart was technically correct but visually dominated by the fastest growers. Switching to a log axis aligns vertical distance with percentage growth, while separated direct labels reduce clutter. The redesign reveals more of the same data; it does not manufacture a new result.

### Name the principles

Appropriate scale, relative rather than absolute encoding, direct labeling, pre-attentive separation, and reduced non-data ink.

SECTION 7

# Master the 10 slides

For each slide, know its job, the one sentence to say, the number to remember, and the transition to the next idea.

# Slides 1-3: establish the question and testable claims

Speaker: Mirthula M.

### Slide 1 - Historical Stock Analysis

Job: open with the umbrella question. Say: 'This is not a price forecast. We test whether apparent diversification survives closer inspection across technology groups and crisis periods.' Remember: 230,111 ticker-days, 49 tickers, about 20 years.

### Slide 2 - What we analyzed and how

Job: explain one row, data quality, and the Python/R split. Say: 'Because prices mix currencies, we compare returns or index every series to 100 rather than placing raw price levels on one axis.'

### Slide 3 - Three hypotheses

Job: state H1, H2, and H3 before presenting results. Transition: 'Together, H1 tests concentration inside technology, while H2 and H3 test whether geographic diversification weakens under stress.'

**Target tone: slow and foundational. Do not rush the definitions, because every later slide depends on them.**

# Slides 4-7: define the test, then show the three results

Slide 4 belongs to Mirthula; slides 5-7 belong to Karthikeyan M.

### Slide 4 - Groups and periods

Name both technology baskets, the calm baseline, and the three market-stress windows. Define correlation, volatility, and volume in plain language. Hand over after the method is clear.

### Slide 5 - H1

Start with index-to-100. Headline: +549% versus +168%; mean daily return 0.26% versus 0.14%; Welch p = 0.0073. End with the selected-groups boundary.

### Slide 6 - H2

Define correlation as co-movement. Headline: 2020 peaks at 0.539 versus calm 0.282. Say 2008 is directionally supportive after widening the basket, but regional coverage remains narrow.

### Slide 7 - H3

Define average absolute return as movement size regardless of direction. Headline: October 2008 = 4.82%, March 2020 = 4.67%. Mention volume only as supporting evidence.

# Slides 8-10: design lesson, conclusion, and next step

Speaker: Lakshmi Mayuri Kavya N.

### Slide 8 - Before and after

Explain that the calculation did not change. The log y-axis makes proportional growth legible; direct labels and reduced non-data ink improve reading. This is a design lesson, not a new finding.

### Slide 9 - Conclusion

Resolve the opening question: selected technology leadership was concentrated, regional protection weakened most in sudden 2020 stress, and volatility independently confirmed panic periods. State patterns, not causality or forecasting.

### Slide 10 - Next steps

Describe an explorable Power BI/Tableau dashboard with visible filters for ticker group, region, and crisis period. Keep the definitions fixed and the limitations visible.

**Closing sentence: 'Success means a user can reproduce every headline and also see where the evidence becomes uncertain.'**

# Three clean handovers

Use the exact idea, not necessarily the exact words.

### Mirthula to Karthikeyan

'Now that the groups, periods, and measures are clear, Karthikeyan will show whether the three hypotheses are supported by the data.'

### Karthikeyan to Lakshmi

'The three results point to the same overall story, but the way we show that story also matters. Lakshmi will explain the chart redesign and close with our next step.'

### Lakshmi closing

'Our dashboard will make these assumptions explorable rather than hiding them. It will let the user reproduce each headline while keeping unequal coverage and other limits visible.'

**If someone forgets a sentence, return to the one job of the slide: define, show one result, state one limitation, and hand over.**

# Likely questions: data and definitions

Short answers first; add detail only if the panel asks.

**Q: Why not compare raw prices?**

A: They use different currencies and arbitrary per-share scales. Returns and index-to-100 are unitless and more comparable.

**Q: Why use Adjusted Close?**

A: It accounts for splits and dividends, so it is more appropriate for return and growth calculations.

**Q: Is the dataset current?**

A: No. It ends on 20 February 2026, so we describe it as historical.

**Q: Are there missing values?**

A: No missing fields and no duplicate Date-Ticker pairs, but there are 1,706 zero-volume rows and uneven market coverage.

**Q: Why is one row called a ticker-day?**

A: Because the unit of observation is one ticker on one trading date, not one company or one market.

**Q: Why use a calm baseline?**

A: It provides a reference level for comparing crisis correlation, volatility, and volume.

# Likely questions: H1 and statistics

Short answers first; add detail only if the panel asks.

**Q: Does 649 mean a 649% gain?**

A: No. Starting at 100 and ending at 649 means a gain of 549%.

**Q: Why these two technology groups?**

A: They were fixed as seven selected semiconductor/AI-linked names and eight selected broader-tech names to test a narrow, explicit H1.

**Q: What does p = 0.0073 mean?**

A: If the group means were truly equal under the test assumptions, a difference this large or larger would be unlikely. It does not give the probability that H1 is true.

**Q: Why Welch's t-test?**

A: It does not require equal variance and is appropriate when the two groups may have different variance or sample size.

**Q: Does H1 prove the AI boom is narrow?**

A: No. It supports outperformance for these selected baskets and dates. A wider AI ecosystem could produce a different result.

**Q: Could one stock drive the result?**

A: Possibly in part. Equal-weighted group averages reduce market-cap dominance, but they do not eliminate standout-company influence. The linked-view exercise was designed to inspect that.

# Likely questions: H2, H3, and design

Short answers first; add detail only if the panel asks.

**Q: Why is 2020 the strongest H2 result?**

A: Its widened-basket correlation is 0.539 versus 0.282 in calm years, a much clearer increase than 2008's 0.327.

**Q: Why did widening the basket matter?**

A: A region represented by one company can be distorted by company-specific news. Averaging several available stocks gives a broader regional signal.

**Q: Why is 2022 different?**

A: Its correlation and volume were below the calm baseline even though volatility was mildly higher, consistent with a slower, uneven drawdown rather than synchronized panic.

**Q: Does high correlation cause volatility?**

A: The project does not test causality. It observes that these measures rose together in some crisis periods.

**Q: Why use mean absolute return?**

A: It captures the typical size of daily movement without positive and negative returns canceling each other.

**Q: Why use a log axis?**

A: The chart compares proportional growth over a very wide range. Equal vertical distances on the log scale represent equal multiplicative changes.

SECTION 8

# Finish with precision

Strong presentation answers are not only confident; they clearly separate evidence, interpretation, limitation, and future work.

# What we can say and what we must not say

These phrases protect the project from overclaiming.

### Safe claims

The selected chip/AI group outperformed the selected broader-tech group. Regional co-movement was highest in the 2020 window. Panic periods showed large volatility and volume spikes in this dataset.

### Unsafe claims

AI caused all technology gains. International diversification always fails in a crisis. Volume causes crashes. The model predicts the next downturn.

- Say 'supports' rather than 'proves.'
- Say 'selected groups' and 'this dataset' rather than 'the whole market.'
- Say 'descriptive evidence' rather than 'causal effect.'
- Say 'historical through 2026-02-20' rather than 'current.'
- Say 'directionally supported for 2008' rather than treating the H2 result as equally strong across crises.

### Best answer structure

Direct answer -> one number -> method -> limitation. Stop there unless the panel asks for more detail.

# What the Power BI or Tableau dashboard should do

The dashboard is an analytical interface, not merely a prettier report.

- Overview page with the central question and the three headline findings.
- Ticker-group filter for chip/AI versus broader tech, with index-to-100 performance.
- Region and period filters for calm, 2008, 2020, and 2022 comparisons.
- Monthly volatility view based on the long-form, month-binned R output.
- Tooltips that state date window, weighting, normalization, and available ticker count.
- Visible warnings for Paris's one-stock basket, sector-narrow regions, share-count volume, and the removed Kaggle page.

**Success criterion**

A user can reproduce each presentation headline by selecting the documented filters and can also see where the evidence becomes uncertain.

**Possible extension**

Run sensitivity checks with alternative basket definitions, rolling correlations, market-cap weighting, and dollar-volume measures if suitable data become available.

# Numbers worth memorizing

These are the figures most likely to be asked about.

| Topic | Number | Meaning |
|---|---|---|
| Dataset | 230,111 x 8 | Ticker-days by columns |
| Coverage | 49 tickers | 2006-01-02 to 2026-02-20 |
| H1 growth | +549% vs +168% | Chip/AI vs broader tech |
| H1 daily return | 0.26% vs 0.14% | Pooled mean daily returns |
| H1 test | p = 0.0073 | Welch test result |
| H2 calm | 0.282 | Widened-basket correlation |
| H2 2008 | 0.327 | Directionally above calm |
| H2 2020 | 0.539 | Strongest regional co-movement |
| H2 2022 | 0.196 | Below calm |
| H3 peak | 4.82% | October 2008 monthly abs return |
| H3 second | 4.67% | March 2020 monthly abs return |
| Volume | 2.7x / 1.7x / 0.9x | 2008 / 2020 / 2022 vs calm |

**Memory trick**

549-168 for H1; 282-327-539-196 for H2; 4.82-4.67 for H3.

# Plain-language glossary

Use these definitions in the room.

| | |
|---|---|
| **Adjusted Close** | Closing price adjusted for splits and dividends. |
| **Daily return** | Percentage change in adjusted close from the previous trading observation. |
| **Index-to-100** | Rescaling so every series begins at 100 on the same valid date. |
| **Correlation** | A unitless measure of how closely two return series move together. |
| **Volatility** | The typical size or dispersion of returns; higher means more movement. |
| **Absolute return** | Return size without direction; both +5% and -5% become 5%. |
| **Equal weighting** | Giving each ticker the same contribution to a group average. |
| **Confidence interval** | A range showing uncertainty around an estimated mean. |
| **p-value** | How surprising the result would be if the null model were true. |
| **Long-form data** | One observation per row with a Ticker field and a value field. |
| **Log axis** | An axis where equal distance represents equal multiplicative change. |
| **Causality** | Evidence that one factor produces another; this project does not establish it. |

# A practical 9-minute run-through

The class slot is 10 minutes with a hard cutoff. Preserve time for Q&A.

| Speaker | Slides | Focus | Target |
|---|---|---|---|
| Mirthula | 1-4 | Question, data, hypotheses, method | about 3:00 |
| Karthikeyan | 5-7 | H1, H2, H3 evidence | about 3:00 |
| Lakshmi | 8-10 | Design, conclusion, next step | about 2:30 |
| Team | Q&A | Short evidence-first answers | at least 1:00 |

- Round 1: speak with notes and stop after 9 minutes.
- Round 2: remove full notes; keep only the number sheet and handovers.
- Round 3: one teammate interrupts with random Q&A from pages 34-36.
- Round 4: present once without stopping, even if a sentence is forgotten.
- Before class: confirm the PPTX opens, charts are legible, and every member knows the first and last sentence of their section.

**If time is running out**

Keep the headline number, the conclusion, and one limitation. Cut extra explanation before cutting the central claim.

# Where every part of this handbook came from

The guide follows the current 10-slide submission and the repository's reproducible outputs.

- Current deck: half-way-presentation/docs/submissions/26120004_2_Presentation.pptx
- Shared data: data/stock_data.csv
- Python hypotheses and charts: assignment_3/analysis.ipynb and assignment_3/figures/
- R replication: assignment_4/analysis.R and assignment_4/docs/submissions/
- R wrangling and monthly volatility: assignment_5/part_2/analysis.R, tables/, figures/, and submission PDF
- Statistical test, design redesign, and widened regions: extra_credit_1/analysis.ipynb, figures/, and submission PDF
- Assignment brief: half-way-presentation/docs/assignment/Project_Status_Presentation.pdf

### Original dataset citation

Ibrahim Shahrukh, Top 50 Companies Dataset, Kaggle: https://www.kaggle.com/datasets/ibrahimshahrukh/top-50-companies-dataset. The source page has since been removed by its uploader.

Related work cited in the presentation notes: Preis et al. (2012), Quantifying the Behavior of Stock Correlations Under Market Stress; Morningstar (2026), 3 Years of the AI Stock Market Boom in Charts; J.P. Morgan (2026), Is It All One Big AI Trade?; Federal Reserve History, The Great Recession and Its Aftermath.

### Final reminder

Know the central story, but keep the limits attached. A careful, qualified answer is stronger than a broad claim the dataset cannot support.